

- Q.25 What is the effect of higher frequencies on measurement using multimeter. (CO3)
- Q.26 Describe De Sauty's bridge. (CO6)
- Q.27 Explain the working principle of a Tong Tester. (CO6)
- Q.28 Describe the working principle of ramp type digital voltmeter. (CO6)
- Q.29 Explain working principle of Q meter. (CO6)
- Q.30 Explain duty cycle of a pulse signal. (CO5)
- Q.31 Explain working of wave analyzer. (CO4)
- Q.32 How a Logic Probe is useful in digital circuits? (CO6)
- Q.33 What are indicating type instruments? Give one example indicating type instrument. (CO2)
- Q.34 Explain limiting error in an instrument. (CO1)
- Q.35 What is a LCR meter give its application. (CO4)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 a) Compare digital and analog instruments. (CO5)
b) Explain the working of a Digital Multimeter with the help of a block diagram.
- Q.37 Draw and explain the different blocks of CRO. (CO6)
- Q.38 Explain the working principle and construction of PMMC type of instruments. (CO3)

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Roll No.

4th Semester/ Mechatronics Subject: Electrical & Electronic Instrumentation & Measurement

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The focusing method used in CRO is (CO4)
a) Mechanical focusing
b) Thermal focusing
c) Electrostatic focusing
d) Electromagnetic focusing
- Q.2 Inductance is measured by which one of the following. (CO6)
a) Wheat stone bridge
b) Schering bridge
c) De Sauty's bridge
d) Hay's bridge
- Q.3 The principle of Q-Meter is based on (CO6)
a) Parallel resonance
b) Series resonance
c) Both A & B
d) None of the above

- Q.4 Lissajous patterns can be used to measure (CO4)
 a) Frequency b) Phase difference
 c) Both A & B d) Intensity of light
- Q.5 The main specification of a CRO is its (CO4)
 a) Weight b) Power supply
 c) Dimensions d) Bandwidth
- Q.6 A moving iron instrument can be used for (CO3)
 a) AC only b) DC only
 c) Both AC and DC d) None of the above
- Q.7 A simple instrument used to detect logic state of a node in digital circuit is known as _____. (CO6)
 a) Logic pulser b) Logic analyser
 c) Logic probe d) Logic comparator
- Q.8 The wave shape generated by time base circuit of a CRO is (CO4)
 a) Triangular wave b) Square wave
 c) Saw tooth wave d) Sine wave
- Q.9 DC bridges are used for measurement of (CO6)
 a) Resistance b) Capacitance
 c) Current d) Inductance
- Q.10 Which of the following is not an integrating instrument? (CO2)
 a) Odometer
 b) Wattmeter
 c) Energy Meter
 d) Water consumption meter

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 The errors caused by human mistakes are known as _____ (CO1)
- Q.12 The input resistance of an ideal ammeter is infinite. (True/False) (CO3)
- Q.13 The loading effect is a desirable feature in measuring instruments. (True/False) (Co1)
- Q.14 Name the instrument used to generate sine waves with variable frequency and amplitude _____. (CO5)
- Q.15 Megger is used to measure _____. (CO6)
- Q.16 Write one application of Spectrum Analyser. (CO6)
- Q.17 RF Stands for _____. (CO5)
- Q.18 Give one example of recording type instrument. (CO2)
- Q.19 Torque to weight ratio of PMMC instrument is high. (True/False) (CO2)
- Q.20 Q-Factor (Quality Factor) of a coil = _____ (Give the formula) (CO5)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write four applications of CRO. (CO4)
- Q.22 Explain the importance of standards in measurement. (CO1)
- Q.23 Elaborate the need of ADC and DAC in a DSO. (CO4)
- Q.24 Explain the main constituents of a low frequency generator? (CO5)